

Data Flow Diagrams and User Stories

Digitize Metro Ticket Booking Using ServiceNow

Date	2 August 2026	Team ID	Not Specified
Project Name	Digitize Metro Ticket Booking Using ServiceNow	Maximum Marks	4 Marks

Context-Level Data Flow

Passenger → Booking Catalog Form → ServiceNow Booking Workflow → Booking Database → QR Ticket S Notification → Passenger

Level-1 Data Flow

Step	Input	Process	Output
1. Request	Station and journey details	Passenger submits catalog request	Validated booking request
2. Validation	Booking request	Check mandatory fields and route validity	Valid request / validation message
3. Fare	Valid request	Apply configured fare rules	Calculated fare
4. Record	Journey + fare	Create booking record and reference	Confirmed booking record
5. Ticket	Booking reference	Generate QR token / digital ticket	QR-code ticket
6. Notify	Ticket + booking details	Send automated confirmation	Passenger notification
7. Monitor	Booking records	Aggregate operational data	Dashboard and reports

User Stories

ID	As a...	I want to...	So that...
US-01	Metro passenger	select source and destination stations	I can book my journey digitally.
US-02	Metro passenger	see the calculated fare before confirmation	I know the exact amount for my journey.
US-03	Metro passenger	receive a QR-code ticket immediately	I can access my ticket without carrying a paper ticket.

US-04	Metro passenger	receive a booking confirmation notification	I have proof of my confirmed booking.
US-05	Metro passenger	view my booking status	I can check whether my ticket is confirmed or used.
US-06	Metro passenger	cancel an eligible booking	I can manage my travel plans digitally.
US-07	Administrator	manage stations and fare rules	The booking process uses current configuration.
US-08	Operations team	view booking trends and failures	I can monitor service performance and demand.

Proposed Flow Diagram Description

Passenger submits request → Flow Designer validates data → Fare calculation action determines price → Booking record is created → QR ticket is generated → Notification is sent → Dashboard receives booking metrics.

Exception Flows

Invalid station selection: return validation message. Fare rule unavailable: place request in exception state for support review. QR generation failure: retain booking record and retry or notify support. Notification failure: preserve ticket in the booking record for manual access.

Benefits:

- Clearly illustrates the movement of data.
- Improves understanding of system processes.
- Simplifies communication among stakeholders.
- Helps identify missing processes or data.
- Supports system design and documentation.

Conclusion

The Data Flow Diagrams provide a clear visualization of how booking information flows through the Metro Ticket Booking System, from the passenger's request to QR code generation and notification. The User Stories capture the functional requirements of application is designed around real user needs. Together, DFDs and User Stories serve as passengers, metro operators, administrators, and system administrators, ensuring that the essential artifacts for system analysis, design, development, and testing, contributing to the successful implementation of the Metro Ticket Booking System using the ServiceNow platform.

